C# Programming Lab Manual

Corrected Copy-Paste Version

Based on the uploaded lab manual. Obvious OCR/typing/syntax errors have been corrected while retaining the source assignment topics and structure.

Important: Console programs are complete programs. Windows Forms entries are event-handler/code snippets and require the corresponding controls in the form.

Assignment 1 – Print Given String Multiple Times

```
using System;

class Program
{
    static void Main(string[] args)
    {
        Console.Write("Enter the string: ");
        string s = Console.ReadLine();

        Console.Write("How many times to print: ");
        int j = Convert.ToInt32(Console.ReadLine());

        for (int i = 1; i <= j; i++)
        {
            Console.WriteLine(s);
        }

        Console.ReadKey();
    }
}
```

Assignment 2 – Different Operators

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int a = 10, b = 20;
        int number = 10;
        bool result;

        Console.WriteLine("a = " + a);
        Console.WriteLine("b = " + b);

        Console.WriteLine("\n1. Arithmetic Operators");
        Console.WriteLine("(a + b) = " + (a + b));
        Console.WriteLine("(a - b) = " + (a - b));
        Console.WriteLine("(a / b) = " + (a / b));
        Console.WriteLine("(a * b) = " + (a * b));
        Console.WriteLine("(a % b) = " + (a % b));

        Console.WriteLine("\n2. Relational Operators");
        Console.WriteLine("(a == b) = " + (a == b));
        Console.WriteLine("(a < b) = " + (a < b));
        Console.WriteLine("(a > b) = " + (a > b));
        Console.WriteLine("(a >= b) = " + (a >= b));
        Console.WriteLine("(a <= b) = " + (a <= b));
        Console.WriteLine("(a != b) = " + (a != b));

        Console.WriteLine("\n3. Logical Operators");
        result = (a == b) || (a > b);
        Console.WriteLine("(a == b) || (a > b) = " + result);

        result = (a == b) && (a > b);
        Console.WriteLine("(a == b) && (a > b) = " + result);

        Console.WriteLine("\n4. Bitwise Operators");
        Console.WriteLine("(a & b) = " + (a & b));
        Console.WriteLine("(a | b) = " + (a | b));
        Console.WriteLine("(a ^ b) = " + (a ^ b));
        Console.WriteLine("(a << 2) = " + (a << 2));
        Console.WriteLine("(a >> 2) = " + (a >> 2));

        Console.WriteLine("\n5. Ternary Operator");
        Console.WriteLine("{0} is {1}", number,
            (number % 2 == 0) ? "Even Number" : "Odd Number");

        Console.ReadLine();
    }
}
```

Assignment 3(a) – Multiplication Table Using For Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int n;

        Console.Write("Enter the number: ");
        n = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Multiplication Table:");

        for (int i = 1; i <= 10; i++)
        {
            Console.WriteLine(n + " x " + i + " = " + (n * i));
        }

        Console.ReadLine();
    }
}
```

Assignment 3(b) – Nested For Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        for (int i = 1; i <= 5; i++)
        {
            for (int j = 1; j <= i; j++)
            {
                Console.Write(j);
            }

            Console.WriteLine();
        }

        Console.ReadKey();
    }
}
```

Assignment 3(c) – While Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int a = 10;

        while (a < 20)
        {
            Console.WriteLine("value of a: {0}", a);
            a++;
        }

        Console.ReadLine();
    }
}
```

Assignment 3(d) – Nested While Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int i = 5;

        while (i >= 1)
        {
            int j = 5;

            while (j >= i)
            {
                Console.Write(j);
                j--;
            }

            i--;
            Console.WriteLine();
        }

        Console.ReadKey();
    }
}
```

Assignment 3(e) – Do-While Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int a = 10;

        do
        {
            Console.WriteLine("value of a: {0}", a);
            a++;
        }
        while (a < 20);

        Console.ReadLine();
    }
}
```

Assignment 3(f) – Nested Do-While Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int i = 0;

        do
        {
            Console.WriteLine("Value of i: {0}", i);

            int j = 0;

            do
            {
                Console.WriteLine("Value of j: {0}", j);
                j++;
            }
            while (j < 2);

            i++;
        }
        while (i < 2);

        Console.ReadKey();
    }
}
```

Assignment 3(g) – Foreach Loop

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int[] arrayInt = { 11, 22, 33, 44 };

        foreach (int m in arrayInt)
        {
            Console.Write(" " + m);
        }

        Console.WriteLine();
        Console.ReadKey();
    }
}
```

Assignment 4(a) – Constructor With Zero Argument

```
using System;

public class Taxi
{
    public bool isInitialized;

    public Taxi()
    {
        isInitialized = true;
    }
}

class TestTaxi
{
    static void Main(string[] args)
    {
        Taxi t = new Taxi();

        Console.WriteLine(t.isInitialized);
        Console.ReadKey();
    }
}
```

Assignment 4(b) – Constructor With Two Arguments

```
using System;

class Rectangle
{
    public int length, width;

    public Rectangle(int x, int y)
    {
        length = x;
        width = y;
    }

    public int RectArea()
    {
        return length * width;
    }
}

class Program
{
    static void Main()
    {
        Rectangle r = new Rectangle(15, 10);
        int a = r.RectArea();

        Console.WriteLine("Area of rectangle is " + a);
        Console.ReadKey();
    }
}
```

Assignment 4(c) – Constructor With Multiple Arguments

```
using System;

class Demo
{
    public Demo()
    {
        Console.WriteLine("\nThe Constructor with no Parameter:");
    }

    public Demo(int a)
    {
        Console.WriteLine("\nConstructor With One Parameter x = " + a);
        Console.WriteLine("Square of x is " + (a * a));
    }

    public Demo(int a, int b)
    {
        Console.WriteLine("\nConstructor With Two Parameters:");
        Console.WriteLine("A = " + a);
        Console.WriteLine("B = " + b);
        int c = a + b;
        Console.WriteLine("Addition is a + b = " + c);
    }

    public Demo(int a, int b, int c)
    {
        Console.WriteLine("\nConstructor With Three Parameters:");
        Console.WriteLine("A = " + a);
        Console.WriteLine("B = " + b);
        Console.WriteLine("C = " + c);
        int d = a + b + c;
        Console.WriteLine("Addition is a + b + c = " + d);
    }
}

class Program
{
    static void Main(string[] args)
    {
        Demo d1 = new Demo();
        Demo d2 = new Demo(10);
        Demo d3 = new Demo(10, 20);
        Demo d4 = new Demo(10, 20, 30);

        Console.WriteLine("Press any key....");
        Console.ReadKey();
    }
}
```

Assignment 5(a) – Single Inheritance

```
using System;

class Item
{
    public void Company()
    {
        Console.WriteLine("Item code = XXX");
    }
}

class Fan : Item
{
    public void Model()
    {
        Console.WriteLine("Fan model = Class");
    }
}

class Program
{
    static void Main(string[] args)
    {
        Item i = new Item();
        Fan f = new Fan();

        i.Company();
        f.Company();
        f.Model();

        Console.ReadKey();
    }
}
```

Assignment 5(b) – Multilevel Inheritance

```
using System;

class StudScience
{
    public void Marks()
    {
        Console.WriteLine("Marks of Student is: 60");
    }
}

class StudCommerce : StudScience
{
    public void RollNo()
    {
        Console.WriteLine("Rollno of student is: 31");
    }
}

class Stud : StudCommerce
{
    public void Percentage()
    {
        Console.WriteLine("Percentage of student is: 70%");
    }
}

class Program
{
    static void Main(string[] args)
    {
        Stud art = new Stud();

        art.Marks();
        art.RollNo();
        art.Percentage();

        Console.ReadKey();
    }
}
```

Assignment 5(c) – Multiple Interface Inheritance

```
using System;

interface Info
{
    void GetInfo();
}

interface Marks
{
    void GetMarks();
}

class Student : Info, Marks
{
    int rno, sub1, sub2;
    string name;

    public void GetInfo()
    {
        Console.Write("Enter the Roll No: ");
        rno = int.Parse(Console.ReadLine());

        Console.Write("Enter the Name: ");
        name = Console.ReadLine();
    }

    public void GetMarks()
    {
        Console.Write("Enter the Marks of Subject 1: ");
        sub1 = int.Parse(Console.ReadLine());

        Console.Write("Enter the Marks of Subject 2: ");
        sub2 = int.Parse(Console.ReadLine());
    }

    public void Show()
    {
        Console.WriteLine("Roll No is: " + rno);
        Console.WriteLine("Name is: " + name);
        Console.WriteLine("Subject 1 is: " + sub1);
        Console.WriteLine("Subject 2 is: " + sub2);
    }
}

class Program
{
    static void Main(string[] args)
    {
        Student d = new Student();

        d.GetInfo();
        d.GetMarks();
        d.Show();

        Console.ReadKey();
    }
}
```

Assignment 6 – Boxing and Unboxing

```
using System;

class Boxing
{
    public void DoBoxing()
    {
        int a = 10;
        object obj1 = a;

        string str = "biral";
        object obj = str;

        Console.WriteLine("BOXING");
        Console.WriteLine("Value type = {0}", obj1);
        Console.WriteLine("Object type value = {0}", obj);
    }
}

class Unboxing
{
    public void DoUnboxing()
    {
        Console.WriteLine("UNBOXING");

        int i = 50;
        object o = 30;

        i = (int)o;
        Console.WriteLine("Value i = {0}", i);

        int b = 100;
        object obj2 = 125;

        b = (int)obj2;
        Console.WriteLine("Final b = {0}", b);
    }
}

class Program
{
    static void Main(string[] args)
    {
        Boxing b = new Boxing();
        b.DoBoxing();

        Unboxing u = new Unboxing();
        u.DoUnboxing();

        Console.WriteLine("Press any key...");
        Console.ReadKey();
    }
}
```

Assignment 7 – Property and Indexer

```
using System;

class Indexing
{
    private string[] range = new string[5];

    public string this[int index]
    {
        get
        {
            return range[index];
        }
        set
        {
            range[index] = value;
        }
    }
}

class Property
{
    private int x;

    public Property(int i)
    {
        x = i;
    }

    public int Z
    {
        get
        {
            return x;
        }
        set
        {
            x = value;
        }
    }
}

class Program
{
    static void Main(string[] args)
    {
        Indexing id = new Indexing();

        Console.WriteLine("INDEXING");
        Console.WriteLine("----------------");

        id[0] = "Gaurav";
        id[1] = "Vikas";
        id[2] = "Ashish";
        id[3] = "Prabhat";
        id[4] = "Vickey";

        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine(id[i]);
        }

        Console.WriteLine("\nPROPERTY");
        Console.WriteLine("----------------");

        Property p = new Property(40);
        Console.WriteLine("p.Z = " + p.Z);

        Console.ReadKey();
    }
}
```

Assignment 8(a) – Exception Handling Using One Catch Block

```
using System;

class Error3
{
    public static void Main(string[] args)
    {
        int a = 10;
        int b = 10;
        int c = 5;
        int x, y;

        try
        {
            x = a / b;
            y = (b + c) / x;
        }
        catch (Exception e)
        {
            Console.WriteLine("Error: " + e.Message);
            y = 0;
        }

        Console.WriteLine("y = " + y);
        Console.ReadKey();
    }
}
```

Assignment 8(b) – Multiple Catch Blocks

```
using System;

class Error4
{
    public static void Main(string[] args)
    {
        int[] a = { 5, 10 };

        try
        {
            int x = a[2] / a[1];
            Console.WriteLine("x = " + x);
        }
        catch (ArithmeticException)
        {
            Console.WriteLine("Division by zero");
        }
        catch (ArrayTypeMismatchException)
        {
            Console.WriteLine("Wrong data type");
        }
        catch (IndexOutOfRangeException)
        {
            Console.WriteLine("ARRAY INDEX ERROR");
        }

        Console.ReadKey();
    }
}
```

Assignment 8(c) – Exception Handling Using Throw

```
using System;

class Program
{
    static void Main(string[] args)
    {
        int a, b, c;

        Console.Write("Enter the first value => ");
        a = int.Parse(Console.ReadLine());

        Console.Write("Enter the second value => ");
        b = int.Parse(Console.ReadLine());

        try
        {
            if (b == 0)
            {
                throw new Exception("You cannot divide by zero");
            }

            c = a / b;
            Console.WriteLine("Result is => " + c);
        }
        catch (Exception obj)
        {
            Console.WriteLine(obj.Message);
            Console.WriteLine("Sorry, you entered wrong value");
        }

        Console.ReadKey();
    }
}
```

Assignment 9(a) – Change Label Text Using Button

```
private void button1_Click(object sender, EventArgs e)
{
    label1.Text = "Hello";
}
```

Assignment 9(b) – Change Label Background Color Using Button

```
private void button1_Click(object sender, EventArgs e)
{
    label1.BackColor = Color.Red;
}
```

Assignment 9(c) – Change Label/TextBox Text and BackColor

```
private void button1_Click(object sender, EventArgs e)
{
    label1.Text = "Hello";
    textBox1.Text = "Hello";
    label1.BackColor = Color.Red;
    textBox1.BackColor = Color.Yellow;
}
```

Assignment 10(a) – ListBox Control

```
// Form Load
private void Form1_Load(object sender, EventArgs e)
{
    listBox1.Items.Add("A");
    listBox1.Items.Add("B");
}

// Button 1: Show selected item
private void button1_Click(object sender, EventArgs e)
{
    string value = listBox1.Text;
    MessageBox.Show(value);
}

// Button 2: Add item from TextBox
private void button2_Click(object sender, EventArgs e)
{
    listBox1.Items.Add(textBox1.Text);
    textBox1.Text = "";
}

// Button 3: Remove selected item
private void button3_Click(object sender, EventArgs e)
{
    int n = listBox1.SelectedIndex;

    if (n >= 0)
        listBox1.Items.RemoveAt(n);
}

// Button 4: Change background color
private void button4_Click(object sender, EventArgs e)
{
    listBox1.BackColor = Color.Red;
}

// Button 5: Exit application
private void button5_Click(object sender, EventArgs e)
{
    Application.Exit();
}
```

Assignment 10(b) – ComboBox Control

```
// Button 1: Show selected item
private void button1_Click(object sender, EventArgs e)
{
    string value = comboBox1.Text;
    MessageBox.Show(value);
}

// Button 2: Add item from TextBox
private void button2_Click(object sender, EventArgs e)
{
    comboBox1.Items.Add(textBox1.Text);
    textBox1.Text = "";
}

// Button 3: Remove selected item
private void button3_Click(object sender, EventArgs e)
{
    int n = comboBox1.SelectedIndex;

    if (n >= 0)
        comboBox1.Items.RemoveAt(n);
}

// Button 4: Change background color
private void button4_Click(object sender, EventArgs e)
{
    comboBox1.BackColor = Color.Yellow;
}

// Form Load
private void Form1_Load(object sender, EventArgs e)
{
    comboBox1.Items.Add("A");
    comboBox1.Items.Add("B");
}
```

Assignment 11 – Show Date and Time Using Timer

```
public partial class Form1 : Form
{
    int second = 0;

    public Form1()
    {
        InitializeComponent();
    }

    private void Form1_Load(object sender, EventArgs e)
    {
        timer1.Interval = 1000;
        timer1.Start();
    }

    private void timer1_Tick(object sender, EventArgs e)
    {
        label1.Text = DateTime.Now.ToString();

        second = second + 1;

        if (second >= 10)
        {
            timer1.Stop();
            MessageBox.Show("Exiting form timer");
        }
    }
}
```

Assignment 12(a) – PictureBox Control

```
private void Form1_Load(object sender, EventArgs e)
{
    pictureBox1.Image = Image.FromFile(
        "C:\\Users\\Public\\Pictures\\Sample Pictures\\Chrysanthemum.jpg");

    pictureBox1.SizeMode = PictureBoxSizeMode.StretchImage;
    pictureBox1.Width = 400;
    pictureBox1.Height = 400;
    pictureBox1.Dock = DockStyle.Left;
}
```

Assignment 12(b) – ScrollBar Control

```
public partial class Form1 : Form
{
    int r, g, b;

    public Form1()
    {
        InitializeComponent();
    }

    private void vScrollBar1_Scroll(object sender, ScrollEventArgs e)
    {
        r = vScrollBar1.Value;
        textBox1.BackColor = Color.FromArgb(r, g, b);
    }

    private void vScrollBar2_Scroll(object sender, ScrollEventArgs e)
    {
        g = vScrollBar2.Value;
        textBox1.BackColor = Color.FromArgb(r, g, b);
    }

    private void vScrollBar3_Scroll(object sender, ScrollEventArgs e)
    {
        b = vScrollBar3.Value;
        textBox1.BackColor = Color.FromArgb(r, g, b);
    }
}
```

Assignment 12(c) – ScrollBar and PictureBox

```
public partial class Form1 : Form
{
    int r, g, b;

    public Form1()
    {
        InitializeComponent();
    }

    private void Form1_Load(object sender, EventArgs e)
    {
        pictureBox1.Image = Image.FromFile(
            "C:\\Users\\Public\\Pictures\\Sample Pictures\\Chrysanthemum.jpg");

        pictureBox1.SizeMode = PictureBoxSizeMode.StretchImage;
        pictureBox1.Width = 240;
        pictureBox1.Height = 240;
        pictureBox1.Dock = DockStyle.Left;
    }

    private void vScrollBar1_Scroll(object sender, ScrollEventArgs e)
    {
        r = vScrollBar1.Value;
        pictureBox1.BackColor = Color.FromArgb(r, g, b);
    }

    private void vScrollBar2_Scroll(object sender, ScrollEventArgs e)
    {
        g = vScrollBar2.Value;
        pictureBox1.BackColor = Color.FromArgb(r, g, b);
    }

    private void vScrollBar3_Scroll(object sender, ScrollEventArgs e)
    {
        b = vScrollBar3.Value;
        pictureBox1.BackColor = Color.FromArgb(r, g, b);
    }
}
```

Assignment 13 – Database Connectivity Using Wizard (Steps Only)

1. Start Visual Studio.
2. Create a Windows Forms Application.
3. Create a Microsoft Access database file.
4. Open Microsoft Office Access.
5. Create a Blank Database.
6. Create a table using Table Design.
7. Enter the required fields and data.
8. Save the database file.
9. In Visual Studio, add a DataGridView control.
10. Click "Add Project Data Source...".
11. Choose "Database" as the data source type.
12. Select the Access database file created earlier.
13. Select the required database objects/table.
14. Finish the Data Source Wizard.
15. The selected data can then be displayed using the DataGridView.